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Acquisition of Japanese Word Descriptions from World Wide Web

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Abstract: In this paper, we propose a method for describing a Japanese word, not with explaining or defining sentences, but with figurative descriptions. Utilizing a simile pattern, our method gathers a large number of noun-noun relations from the World Wide Web. On the basis of those relations and their statistical information, associative pieces of knowledge called descriptors are estimated. The descriptors, which describe a query word figuratively, are sorted by ranking in order of descriptive ability level with generality and locality. As output, a set of sorted descriptors is shown with several types of output forms.

Some experiments using a prototype system have been conducted. The experimental results show that the fundamental performance of our method is significantly better than the bag-of-words approach. Also, the responsiveness of the method for hot keywords on information retrieval web sites shows that the outcome of the evaluation had 60 % precision, which exceeds that of a common dictionary. Furthermore, it is possible that the proposed method could be comparable to Wikipedia if steady coverage of the figurative descriptions for a query word could be ensured.

Key words: figurative descriptions, Japanese word, WWW, stereotyped pattern

1. Introduction

When we encounter an unknown word, sentences describing the word help us to understand it. For example, an unknown term "*Randy Johnson*" is described by the following sentences[1].

Example of Descriptive Sentences

Randall David Johnson (born September 10, 1963), nicknamed "The Big Unit," is a former left-handed Major League Baseball starting pitcher. Over a 22-year career, Johnson played for six different teams. The 6-foot-10-inch (2.08 m) Johnson was celebrated *for having one of the most dominant fastballs in the game. He regularly approached, and occasionally exceeded*, 100 miles per hour (160 km/h) during his prime. He also threw a hard, biting slider. ...

Giving answering sentences to a user's query, some kinds of higher-level question answering (QA) methods, which definitional-type QA, why-type QA and how-type QA, are being investigated[1][2][3]. To deal with such complex tasks, it is indispensable not only to apply a factoid-type question answering method but also to combine deep analysis, context analysis, summarization and so forth. Although it is possible to secure superior performance by a combination of such complex techniques, it remains difficult. Moreover, it's inefficient. At any rate, it is important to collect enough elements or components to generate sentences.

Consequently, we focus on providing enough elements to imagine the word sense instead of answering without exaggeration and without omission about a query word. It is assumed that pieces of knowledge called *descriptors* stimulate a user's association about a word sense as effectively as a sticky set.

Example of Ideal Figurative Descriptions

Randy Johnson
- **Position:**
  starting pitcher
  left-handed pitcher
- **Affiliation:**
  Major League Baseball
  Arizona Diamondbacks
- **Characteristic:**
  6-foot-10
  a tall man
- **Ability:**
  biting slider
  fastballs
  100-mph pitches

  ⋮

We put forward a method that gleans and coordinates figurative descriptions of a query word from the WWW. First of all, applying a query word to a stylized context pattern, a figurative expression is generated. Second, descriptive words related to the query word are extracted through information retrieval about the figurative expression from the WWW. Then, the descriptive words are ranked and integrated by scores based on frequencies. Depending on a user's choice, a result is visualized as several forms. Utilizing a set of knowledge pieces in the description, the user can imagine the query word and understand the sense of the word by association.

Hereafter, the methodology of our method will be illustrated in section 2. The experiments performed to evaluate the performance of the system are described in section 3, and the experimental results are discussed in section 4.

[1] *http : //en.wikipedia.org/wiki/Randy_Johnson*

Table 1. Some Examples of Figurative Expression

Form	Example
source +indicator +target	ランディ・ジョンソンのような投手 (pitcher like Randy Johnson)
source +indicator +target	ランディ・ジョンソンのような剛速球 (dominant fastball like Randy Johnson)
source +indicator +target	ランディ・ジョンソンのような高速スライダー (biting slider like Randy Johnson)

2. Role of Figurative Relation and Descriptions

A figurative expression has the function of describing a certain instance with another instance[4],[5]. On the basis of a language form, a figurative expression can be classified into three kinds of elements: source, target, and indicator. A figurative expression can be interpreted by comparing the meaning of a source element and a target element in it[6]. Those figurative relations often appear in practical documents like newspaper articles and web documents. Frequently, not only common nouns but also proper nouns become figurative elements.

Among all figurative expressions, a simile specifies all the elements such as "ランディ・ジョンソンのような速球 (a fastball like Randy Johnson's)". In that respect, figurative indicators can be utilized as stereotyped patterns. Some examples of simile expressions with the "*A* のような *B*(*B like A*)" pattern are shown in Table1. Each simile expression describes a figurative relation or a literal instance. Meanings of those expressions are grasped through comparison of source and target. Interestingly, when interpreting a simile expression, compared elements have other specific relations besides the figurative relation. In the first example, "ランディ・ジョンソン (Randy Johnson)" and "投手 (a pitcher)" have an *is-a* relation. In the second example, "剛速球 (dominant fastball)" and "高速スライダー (biting slider)" have *subject-attribute* relations with "ランディ・ジョンソン".

Considering the above circumstances, it can be imagined that "ランディ・ジョンソンは投手であり，剛速球と高速スライダーを投げる．(Randy Johnson is a pitcher, and he throws a dominant fastball and a biting slider.)" To specify those relations, utilizing indicating patterns as markers is very useful.

Acquisition of knowledge based on a stereotyped pattern has lately attracted considerable attention[7], [8], [9], [10], [11] . Several studies reported that that approach is more effective than traditional methods. There are some fundamental principal investigations of such directions as follows. Hearst estimated how co-occurring word pairs can be collected from corpora utilizing the pattern "*A such as B*"[7]. The followings are the cases of the large-scale approach. Pantel et al. and Shinzato et al. gathered numbers of co-occurrence knowledge from the WWW with several kinds of patterns to construct a large-scale ontology[11], [9]

Masui et al[12] proposed a method to extract property knowledge from the WWW by putting together pattern-based knowledge acquisition and figurative indicators. Their method extracts a concept word x_i with a descriptive word γ_j from an adnominal expression. For example, from an expression "赤いリンゴ (red apple)", "リンゴ (apple)" is regarded as x_i and "赤い (red)" is extracted as γ_j. Applying the above process to large corpora, a set of descriptive words is procured as $\Gamma = \{$"赤い (red)", "甘酸っぱい (sweet and sour)", "大きい (big)", ... $\}$.

Next, expressions such as "リンゴのように赤い (red like an apple)" or "リンゴのように甘酸っぱい (sweet and sour like an apple)" are generated by combining x_i, γ_j and an indicator ω. In this case, ω is set by "のように". Every generated expression above is estimated by WWW-based relevance feedback: "赤い (red)" and "甘酸っぱい (sweet and sour)" are validated as appropriate modifiers although "大きい (big)" is judged to be inappropriate. Thus, candidates for "property" are refined.

We expand the above method to include *noun-noun* relations and achieve effectual descriptor acquisition. As a query word, a noun q_i is given. Combination of q_i and descriptor γ_j in figurative context ω can be expressed as a 3-tuple as (x_i, ω, γ_j), and (X, ω, Γ) means a stylized figurative expression pattern. In the pattern, ω is set by "のような (like a/ as a)". Frequencies of $(q_i, \omega, *)$ and (q_i, ω, γ_j) can be fundamental to estimating the relativity of γ_j against a set of descriptors related to q_i, that is to say, the saliency of γ_j for q_i. The word q_i is applied to X of stereotyped pattern (X, ω, Γ). Retrieving the generated expression $(q_i, \omega, *)$, a set of descriptors $\{\gamma_1, \gamma_2, \ldots, \gamma_j, \ldots\}$ is collected for the wild card slot "*" in the expression. Applying γ_j to the stereotyped pattern (X, ω, Γ), a related expression $(*, \omega, \gamma_j)$ is generated. Retrieving the expression from the WWW, noun q_k related to γ_j can be gathered. q_k is regarded as a synonym for the query word q_i, which is described by γ_j. Therefore, frequencies of $(*, \omega, \gamma)$ and (q_k, ω, γ_j) can be fundamental to estimating the relativity of γ_j against a set of descriptors related to q_k, that is to say, specificity of γ_j for q_k.

Considering the saliency and specificity mentioned above, the power of γ_j to describe a query word q_i can be estimated.

3. Methodology

We implemented our proposed method on a prototype system named *Murasaki* that describes an input query word with Japanese figurative descriptions. Murasaki extracts the relation between two words based on comparative structure. No statistical method can extract word relations like Murasaki does. Moreover, public opinion and topic transition against a query word will be followed flexibly because the system also extracts such a word relation from the WWW dynamically.

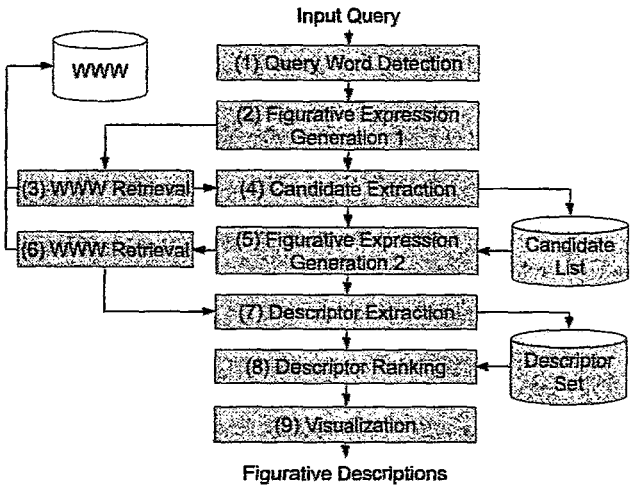


Figure 1. Processing Flow of Murasaki

3.1 Description of Process

The construction of the implemented system is described in Figure 1. The system is composed of ten steps as follows:

1. **Query word detection:** An input query word x_i is recognized and registered. For now, only a noun is accepted.
2. **Figurative expression generation (1):** By applying the query word x_i to the slot X in the stylized pattern (X, ω, Γ) that is set beforehand, a figurative expression is generated. Therefore, an expression $(x_i, \omega, *)$ is generated. As an indicator, "のような (like a/as a)" is set to ω. For example, given a query word "Randy Johnson", it is set to the pattern and an the expression ("ランディ・ジョンソン", ω, $*$) is generated.
3. **WWW retrieval:** Using a WWW search engine, documents relevant to the expression $(x_i, \omega, *)$ are retrieved. Simultaneously, snippets including figurative expressions (x_i, ω, γ_j) are collected from retrieved results. The number of retrieved pages m can be set from 100 to 1,000 by the user.
4. **Candidate extraction:** After all of the collected snippets are processed by morphological analysis, then figurative expressions (x_i, ω, γ_j) are recognized. For now, an element γ_j is defined as a noun, a compound noun and "A の B(B of A)" . Afterward, every γ_j is extracted from the figurative expressionsand registered to a candidate list.
5. **Figurative expression generation (2):** By applying the candidate word γ_j to the slot Γ in the pattern (X, ω, Γ) that is set beforehand, a figurative expression is generated. Therefore, expressions (x_i, ω, γ_j) and $(*, \omega, \gamma_j)$ are generated. For the example of "ランディ・ジョンソン (Randy Johnson)", prepared expressions with candidate "投手 (pitcher)" are ("ランディ・ジョンソン", ω, "投手") and ($*$, ω, "投手"). For the other candidate "剛速球 (dominant fastball)", ("ランディ・ジョンソン", ω, "高速スライダー (biting slider)") and ($*$, ω, "高速スライダー") are generated.
6. **WWW retrieval:** Using a WWW search engine, documents relevant to the expressions are retrieved. First, the number of hit pages related to (x_i, ω, γ_j) is acquired. Second, the number of hit pages related to $(*, \omega, \gamma_j)$ is acquired.
7. **Descriptor extraction:** Significant descriptors are selected from the candidate list based on the number of hit pages. The user can set the significant number of hits to 10^2 (on) or 0 (off).
8. **Descriptor ranking:** Each score for descriptors is computed, and the descriptor set that describes a query word x_i is integrated. The detailed computation and formula are explained below.
9. **Visualization:** Descriptors are ranked in order of each score and read out. As a default, the output is graphed in Figure2. Following each link on a descriptor, a web document includes co-occurrences of the query and descriptors are shown. Alternatively the format can be chosen from a graph, a mapping of the relation between descriptors and a list of snippets highlighting the comparative expressions including the descriptor.

3.2 Ranking Score

Every descriptor is scored with frequency for a word pair of a query word and the descriptor. Each descriptor's score corresponds to a relation between a query word and its descriptor in a figurative expression. In this case, a context pattern "のような (like a/as a)" is used as ω in the expression (q_i, ω, γ_j). The relation between the descriptor and the query word is calculated by formula(1)[12].

$$score(\gamma_j|q_i) = f_\omega(\gamma_j|q_i) * icf_\omega(\gamma_j) \quad (1)$$

Let ω be the figurative context and $f_\omega(\gamma_j|q_i)$ be the frequency count of a descriptor γ_j occurring with a query word x_i in ω. In formula(2) below, $icf_\omega(\gamma_j)$ is defined as the inverted frequency of a query word against a descriptor γ_j. Let $f_\omega(\gamma_j)$ be the frequency of words occurring with a descriptor γ_j in a figurative context ω, and N be the total number of words occurring in the context ω.

$$icf_\omega(\gamma_j) = log\frac{N}{f_\omega(\gamma_j)} + 1 \quad (2)$$

The results are shown as a graph, a list of snippets including each descriptor, and a list of descriptor links to relevant web pages in which each descriptor co-occurs with a query word. One of the output examples and a screenshot are shown in Table2 and Figure2.

4. Experiments

Experiments were performed to evaluate our implemented method.

[2]We decided that 10 is the most suitable number by the preliminary investigation.

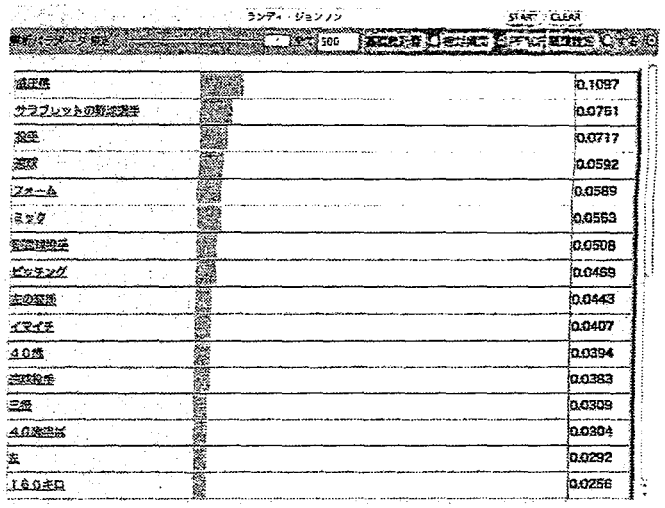


Figure 2. Example of Murasaki's Output

Table 2. Descriptions of "ランディ・ジョンソン (Randy Johnson)"

rank	descriptor	score
1	威圧感 (formidableness)	0110
2	サラブレットの野球選手 (a natural baseball player)	0.076
3	投手 (a pitcher)	0.072
4	速球 (fastball)	0.059
5	フォーム (form)	0.059
6	ミック (a strong-armed pitcher)	0.056
7	剛速球投手 (a strong-armed pitcher)	0.051
8	ピッチング (pitching)	0.047
8	左の変則 (unorthodox left hand)	0.044
10	イマイチ (not the greatest)	0.041
11	４０歳 (40 years old)	0.039
12	速球投手 (fastball pitcher)	0.038
13	三振 (strikeout)	0.031
14	４０歳半ば (mid-forties)	0.030
15	左 (left)	0.029
16	１６０キロ (100 mph)	0.026
17	肩甲骨 (blade bone)	0.022
18	豪腕 (rubber arm)	0.021
19	息 (standing)	0.019
20	２０年以上活躍 (performed well for over 20 years)	0.017

The descriptions above were acquired July 20, 2010.

4.1 Basic Performance

The fundamental performance of the proposed method was evaluated.

First, 20 evaluators judge each descriptor set that the system returned for a query word as appropriate or not. The queries are words occurring to the evaluators. Evaluators are 20 university students 20 to 23 years old. Number of pages n was set as 1,000 to collect descriptor candidates. As a consequence, 173 judgments were obtained. Second, we sampled descriptor sets on 70 query words at random from the 173 judgments and acquisition was repeated 50 times. As an evaluated value, the average precision for all query words was calculated.

According to human judgment, precision values were computed for each sample. Relative recall of our method was calculated against the baseline result[13]. Relative recall shows what percentage of necessary answers a system has output. This measure helps to consider system coverage when it is difficult to calculate recall.

The same evaluation was applied to a baseline system. The baseline does not use a stereotyped pattern but a fixed m-word size window in which a query word is the head. From the head, the nearest neighbor noun ϵ is extracted as a descriptor with its own frequency.

We got baseline-based descriptions by changing the window size m from 2 to 12. The optimum value is regarded as $m = 6$ according to comparison of each result above. Our method was compared with the baseline based on two measures: precision and relative recall.

4.2 Responsiveness to Popular Search Keywords

We evaluated how responsive the Murasaki system returns to recent words.

To begin with, the following keywords were acquired from the principal WWW retrieval sites: (a) 30 words from Yahoo! Japan's hot search keywords, (b) 20 words from Hatena hot keywords, (c) 10 words from hot keywords in goo and (d) 10 words from Google hot keywords[3]. As a consequence, a total of 61 words were listed for evaluation because 9 compound words were out of the system specifications.

Each keyword was input to the Murasaki system as a query word, and returned descriptors were evaluated(number of web pages to be retrieved: $n = 500$). As a control, entries of the hot keyword were confirmed using a common electronic dictionary[4] and Wikipedia.

4.3 Experimental Results

The proposed method scored 0.53 on precision, exceeding the 0.35 average of the baseline systems. Relative recall of the method scored 1.11 against the 6 window-sized baseline system, which is the best of baselines. The obtained results show that our proposed method surpassed the bag-of-words model.

Figure 3 shows the results of responsiveness evaluation. In the graph, performance for each popular web search keywords (a) to (d) and the average scores are plotted. "Murasaki(hit ratio)" shows the ratio of query words for which our system succeeded in collecting their descriptors. "Murasaki(precision)" shows the ratio of query words for which our system succeeded to collecting their appropriate descriptors. "Wikipedia" shows the ratio of query words that are entry words in Wikipedia. "Dictionary" shows the ratio of query words that are entry words in a common dictionary.

As for average value, the proposed method scored 0.60 on precision, and 0.81 on the hit ratio, as did Wikipedia.

[3]In each case, those keywords were registered May 10, 2010.
[4]Digital Dai-Ji-Sen includes 230,000 words

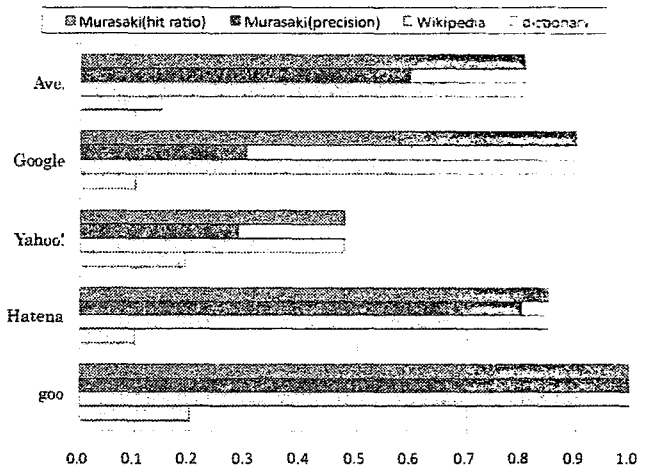


Figure 3. Comparison of Responsiveness to Popular Web Search Keywords

For the common dictionary, precision was 0.15. According to each keyword list, the result for list (d) was the best, and Wikipedia and the proposed method dealt with all entries. On the contrary, for list (a), Wikipedia scored 0.48 on precision, our system scored 0.29, and the common dictionary scored 0.19.

5. Discussion

Below, we discuss the evaluation results. According to precision and relative recall, our proposed method surpassed the bag-of-words model. In addition, we computed MRR (mean reciprocal rank), which is a measure to estimate a system's ranking performance for the results, and the mean value of MRR for the 50 trials was 0.74. This means that the system also showed good ranking performance in that it consistently ranked appropriate descriptors in the top or second-best slots.

Regarding responsiveness, few entries of hot keywords were found in the common dictionary. This is because most hot keywords are proper nouns such as "龍馬伝 (Ryoma-den)", "竹内結子 (Takeuchi Yuko)", and "スリーエフ (Three F)". In contrast, the characteristic that new pages for new words and proper nouns are registered in a short time led Wikipedia to secure good performance.

The evaluation result of the proposed method exceeded that of the common dictionary but did not beat Wikipedia. Therefore, our method is valid for new words and proper nouns for which there are no static measures. Moreover, in almost all difficult cases appropriate descriptors were collected but not enough to describe the query word.

Comparing the hit ratio (for the number of acquired descriptors), the proposed method performed as well as Wikipedia. This suggests that the proposed method has the potential to provide comparable results to Wikipedia, if a sufficient number of descriptors can be acquired. Some words such as "擁立 (enthronement)", "FOIL" and "ルックアップ (look-up)" were not registered in Wikipedia but

Table 3. Examples of Figurative Descriptions(1)

descriptor	score
龍馬伝 (Ryoma-den)	
1年間のドラマ (yearlong drama)	0.271
時代劇番組 (historical play program)	0.123
ウジウジウダウダ (wishy-washy)	0.097
世界 (a world)	0.093
誠実 (honesty)	0.081
ぐるぐるパーマ (rolled-up perm)	0.062
NHK の大河ドラマ (NHK period drama)	0.062
勢い (energy)	0.043
危惧 (misgivings)	0.039
自分 (oneself)	0.039
封建的暗黒社会 (feudalistic underworld)	0.034
骨太のドラマ (a tough-guy drama)	0.028
ドラマ (a drama)	0.028
竹内結子 (Yuko Takeuchi)	
別れ方 (how to get a divoce)	0.303
美人先生 (a beautiful teacher)	0.175
女優 (an actress)	0.153
美人 (a beauty)	0.149
産後ダイエット成功者 (a postpartum dieting winner)	0.039
女性 (a woman)	0.033
美男美女だらけ (full of good-looking people)	0.011
嫁 (wife)	0.011
超弩級の美人 (a super beauty)	0.010
:	:
地震速報 (earthquake early warning)	
情報 (information)	0.225
テロップ (a scrolling ticker)	0.219
客観データ (objective data)	0.097
緊急 (emergency)	0.093
プッシュ型の通信 (push-type communication)	0.077
チャイム (chime)	0.075
リアルタイム情報 (real-time information)	0.043
速報性 (quickness)	0.022
命 (life)	0.021
ピンポン (ding-dong)	0.021
データ (data)	0.019
災害時の緊急速報 (emergency alert in a disaster)	0.017
ニュース的 (news-like)	0.013
警告音 (warning tone)	0.011
:	:
iPad	
端末の市場 (market of terminal unit)	0.264
端末 (terminal unit)	0.198
タブレット (tablet)	0.087
デバイス (device)	0.068
タッチスクリーン式端末 (touch-screen terminal)	0.026
メディアタブレット (media tablet)	0.023
タッチ (touch)	0.023
デバイスの普及 (spread of devices)	0.023
電子書籍 (electronic book)	0.023
情報端末 (information terminal)	0.022
タブレット型 (tablet-type)	0.020
タブレットデバイス (tablet device)	0.015
製品 (products)	0.014
大型のスクリーン (big screen)	0.013
タブ (tab)	0.012
高級ネットブック (luxury netbook)	0.011
機器 (equipment)	0.010
タブレット型端末 (tablet-type terminal)	0.012
:	:

Murasaki could provide descriptions. Because "擁立" is a super-universal word, it was not registered by Wikipedia. "FOIL" and "ルックアップ" are ambiguous words because of several kinds of contracted forms. Furthermore, this can cause Wikipedia to not make an entry for meanings of those words that are not widely known. This means that our method has the potential to process irrespective common nouns and proper nouns.

We also focus on the fact that there are different kinds of descriptors such as broader class, attribute, instance and metaphor. In the experiment, it can be considered that mixing those kinds of descriptors caused conflict in the WWW retrieval. Applying the classified descriptor to expand a query will be effective to retrieve a document for specific purposes.

Table 4. Examples of Figurative Descriptions(2)

descriptor	*score*
紫 (purple)	
色 (color)	0.427
ピンク (pink)	0.115
緑 (green)	0.069
青 (blue)	0.056
茶色 (brown)	0.024
:	:
ペン回し (spinning a pencil)	
回転運動 (rotative motion)	0.220
講座サイト (lecture sites)	0.184
思考上 (conceptional)	0.178
支点 (fulcrum)	0.139
自転 (rotation)	0.124
:	:
ラミノーズテトラ (rummy-nose tetra)	
体型 (shape)	1.000
ベンター博士 (Dr. Venter)	

Table4 shows invalid results. There are two cases: failure to acquire appropriate descriptors though descriptors were collected: and failure to acquire a sufficient number of descriptors. The former case was caused by abstraction of the query word. For example, it is hard to imagine clear descriptions for "紫 (purple)" and "ペン回し (spinning a pencil)". About the queries, many descriptors were synonyms although a sufficient number of descriptors were acquired. The latter resulted from from low acceptance of the query words. "ラミノーズ・テトラ (rummy-nose tetra)" is a term known by a small population of fish enthusiasts or experts. Therefore, it is rarely written in web documents. To deal with those difficult queries, supporting image data is predictably effective.

Moreover, some cases have still problems though the results show total effectiveness. Inappropriate elements are included in the results, and a number of synonymous terms ranked in the top. For instance, "iPad" descriptions are totally effective. But the top rank descriptor "端末の市場 (tablet market)" cannot be recognized as the best. To imagine the query word, it is inefficient for a number of synonyms such as "タブレット (tablet)", "メディア・タブレット (media tablet)" and "タブレットタイプ (tablet type)" to rank in the top of their descriptions. To solve those problems, we can introduce a method to estimate identity between synonymous terms before the ranking process, a method to reduce the number of elements to be classified by re-ranking each classified set, and a method to show descriptors exhaustively.

6. Conclusions

We proposed a method to acquire knowledge to describe a Japanese word figuratively. Using a stylized context pattern and feedback process, figurative knowledge called descriptors is extracted from the WWW. The implemented system was verified by some experiments. Basic performance was evaluated by comparison with baselines, and responsiveness to hot keywords was compared with Wikipedia and a common dictionary.

The results showed that the proposed method was beneficial. There are many avenues of future work for improving our implemented method. Next, we will use our system to classify descriptors based on bootstrapping combination of several kinds of patterns.

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